

Solution Requirements

Date	2 October 2026
Team ID	SWTID-2026-8368
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Not required: single-user local app. The Gemini API key is held server-side in .env and never shown to the user.	Low
2	Authorization levels	One role (the person using the form). No admin or user tiers.	Low
3	External interfaces	Google Gemini API (outline with Flash, story with a second model) and the Hugging Face hub for the one-time Stable Diffusion v1.5 download.	High
4	Transactions processing	A form submit (POST /generate) runs the pipeline: outline, then story and 5 panel images in parallel, then layout and PDF.	High
5	Reporting	Preview page of all panels and a downloadable PDF (one page per panel); JSON result at /generate-comic/json.	High
6	Business rules, etc.	Exactly 5 panels; outline returned as JSON; story in fixed Panel / Caption / Narration format; setting, tone and style come from fixed options.	High
7	Compliance to laws or regulations	No personal data is collected or stored. Respect Gemini free-tier terms and Stable Diffusion v1.5 licence.	Medium

8	Other	Test endpoint /test-image; clear error message when the Gemini quota is exhausted.	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Panel images generated in fp16 on MPS/CUDA with the model warmed up at start; story generated in parallel with images.	About 12.4 s per panel image; full comic in about 104 s (measured)
2	Scalability	Single process on one machine; image generation is the bottleneck.	1 comic at a time on a 16 GB laptop; scale-out plan in Scalability & Future Plan
3	Security & Data Privacy	API keys only in .env (git-ignored); no user data stored; filenames sanitised for saved images.	No secrets in the repository
4	Reliability & Availability	Pipeline errors are caught and returned as readable HTTP 500 messages.	Quota error shown clearly; 24 automated tests pass
5	Usability & Accessibility	Short form with pill options, loading state on the button, responsive layout.	Usable on desktop and mobile widths
6	Other	Maintainable modules, configurable model names, Unicode PDF fonts.	Model names changed through .env without code changes